Rated power at 50 kW

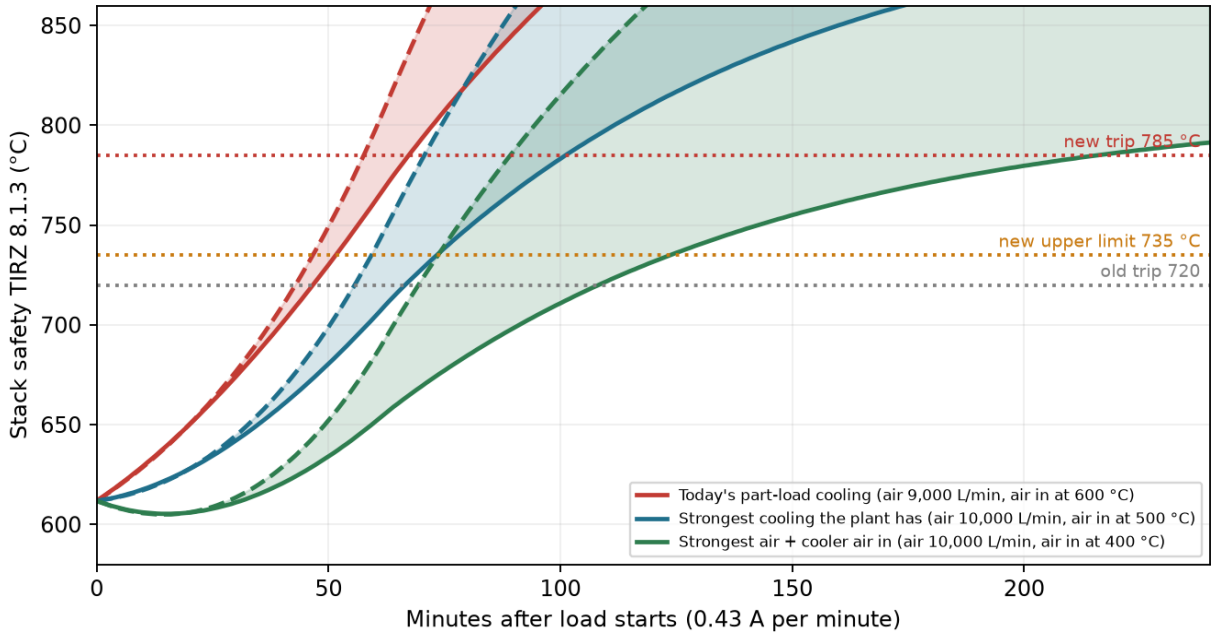
SOFC POC plant: can the rated power stage hold 50 kW with a 735 °C upper limit and a 785 °C trip on the stack safety thermocouple? 2026-11-06. Same content as the online pages.

Contents: 1 The answer · 2 What 50 kW would take · 3 Old limits vs new limits · 4 What can be changed · 5 Plant modifications

1 Can the plant run at 50 kW with a 735 °C limit and a 785 °C trip?

> **Not with the cooling the plant has.** The stack safety temperature passes 735 °C, then the 785 °C trip, and keeps rising. The higher limits only delay the trip: by 15–35 minutes with the plant's own cooling, and by up to 108 minutes with cooler air going in.

50 kW needs about 29.2 A per string (5 strings) and about 166 L/min of natural gas. The plant never ran above 16 A, so the measured behaviour is stretched upwards (same method as the earlier rated-power check).



Each band is one cooling setting; solid edge = twin heat rule, dashed edge = physics rule with extra cell heat.

Cooling	Passes 735 °C after	Trips at 785 °C after	Would settle at
Today's part-load cooling (air 9,000 L/min, air in at 600 °C)	47–52 min	58–68 min	1,088–1,547 °C
Strongest cooling the plant has (air 10,000 L/min, air in at 500 °C)	60–74 min	71–102 min	917–1,216 °C
Strongest air + cooler air in (air 10,000 L/min, air in at 400 °C)	74–124 min	90–217 min	811–1,085 °C

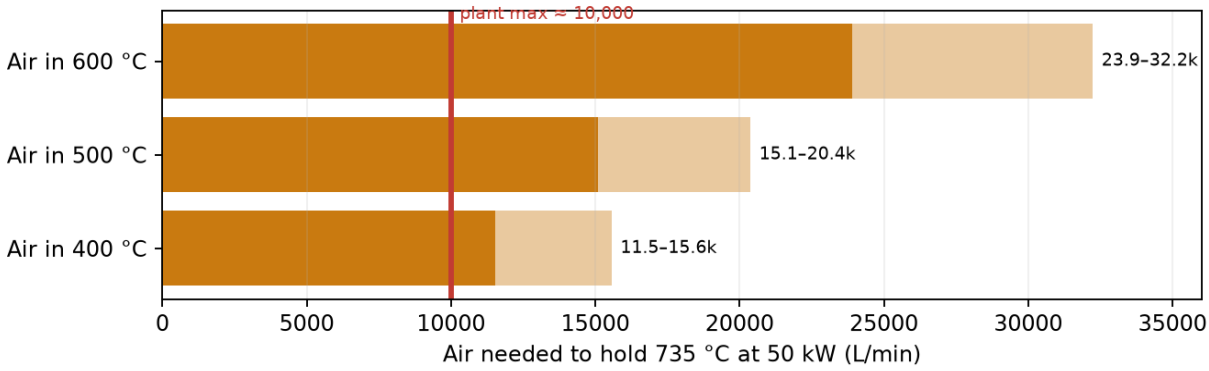
There is no resting point under 785 °C at 50 kW with any cooling the plant can give; the stack would keep heating towards 811–1,547 °C.

If you remember one thing: with today's air supply the plant cannot hold 50 kW; the stack passes 735 °C and trips at 785 °C, just later than before.

2 What 50 kW would take

To hold the stack at 735 °C at 50 kW the plant needs **1.2–3.2 times** its maximum cooling air, depending on the air-in temperature. With today's air, the most it can hold steadily under 735 °C is about **29–33 kW**.

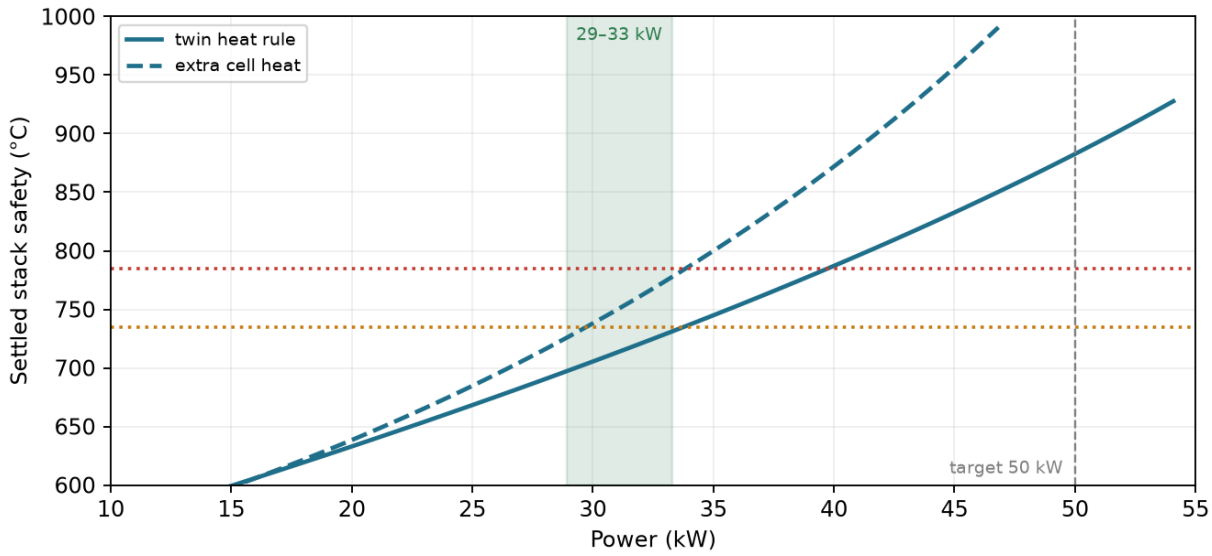
Option 1: more air



Air in	Twin rule (L/min)	Extra cell heat (L/min)	Times plant max
600 °C	23,898	32,243	2.4–3.2 ×
500 °C	15,105	20,380	1.5–2.0 ×
400 °C	11,540	15,570	1.2–1.6 ×

The air rule was fitted only up to 10,000 L/min, so these numbers are a second stretch.

Option 2: accept a lower power

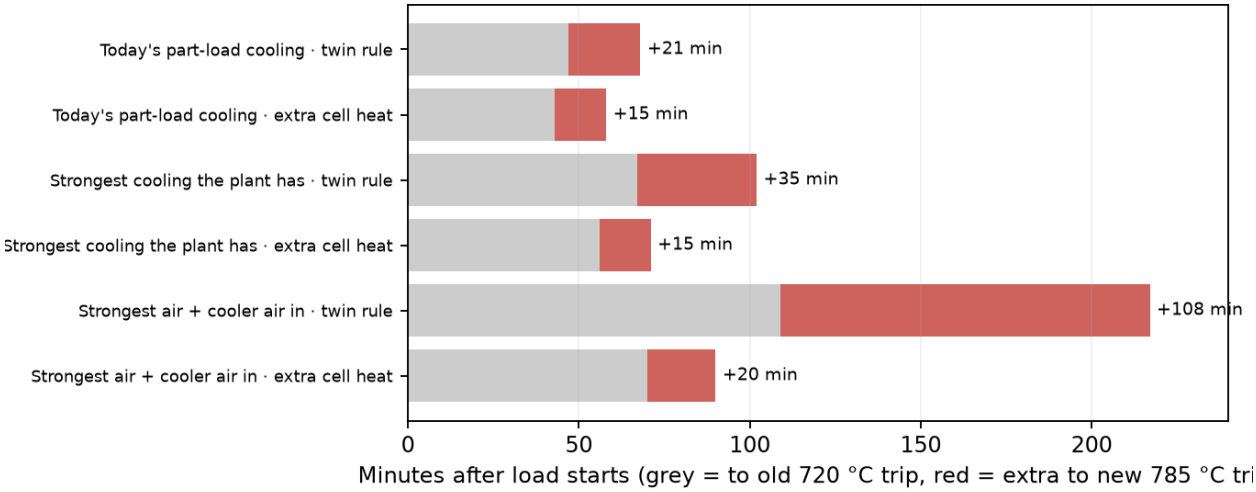


Settled stack safety temperature at each power with the strongest cooling (10,000 L/min, air in 500 °C). It crosses 735 °C at about 29–33 kW.

If you remember one thing: 50 kW under 735 °C needs a much bigger air supply or much cooler air going in; today's plant holds only about 29–33 kW.

3 Old limits vs new limits

Moving the trip from 720 °C to 785 °C gives **15–35 extra minutes** with the plant's own cooling, up to 108 minutes with cooler air going in. It does not make 50 kW steady.



	Old	New (rated power stage)
Upper limit held	stack ≈ 660 °C, safety < ~700 °C	safety 735 °C (stack ≈ 701 °C)
Safety trip TIRZ 8.1.3	720 °C	785 °C
Highest steady power*	about 26–29 kW	about 29–33 kW
50 kW steady?	no	no

* Strongest cooling the plant has (10,000 L/min, air in 500 °C).

Exhaust: the exhaust outlet safety TIRZ 8.1.10 trips at 300 °C since 27 March 2026 and reaches it only after the stack reaches 785 °C, so the stack trip stops the plant first. The exhaust estimate is rough (±24 °C).

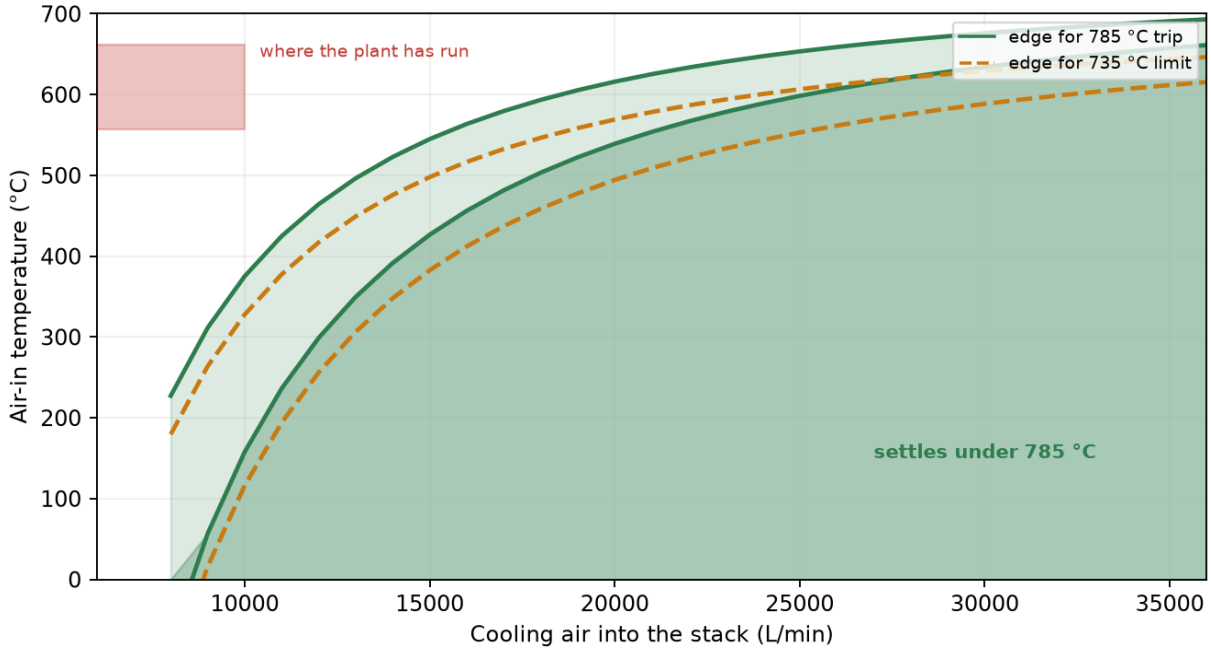
Trust: sure that more current means more heat and the air is near its maximum; likely that 50 kW cannot be held under 735 °C with today's air (both heat rules agree, large gap); rough on exact minutes (stretched beyond 16 A). Not covered: stack maker's approval of 735/785 °C and the 0.72 V cell-voltage limit that already stops current near 15 A.

Decisions: (1) add a Rated power stage with 735/785 °C to the digital twin? (2) keep 50 kW (bigger blower, cooler air in) or set the stage to about 29–33 kW? (3) confirm 735/785 °C with the stack maker.

If you remember one thing: the new 735 / 785 °C limits only buy time; to hold 50 kW the plant needs more cooling, not higher limits.

4 What can be changed so it settles under the trip?

With today's plant only **lower power** works: about **33–39 kW** settles under 785 °C, 29–33 kW under 735 °C. Keeping 50 kW needs about 2–3 times today's cooling air, or more air plus cooler air going in.



At 50 kW, combinations in the green area settle under 785 °C (darker = both heat rules agree). Red box = where the plant has run in part load.

Change (at 50 kW)	Under 785 °C	Under 735 °C	Verdict
1 Lower power (air 10k, air in 500 °C)	up to 33–39 kW	up to 29–33 kW	works today
2 More air (air in 600 °C)	18,531–25,180 L/min	23,898–32,243 L/min	new blower 1.9–3.2 ×
3 Cooler air in (air 10k)	158–375 °C	116–328 °C	outside stack experience
4 Air 15k + cooler air in	air in 427–545 °C	air in 383–498 °C	blower + stack maker OK
5 Less natural gas	settles 822–1,098 °C even at zero		no

Ranges: low end = twin heat rule, high end = physics rule with extra cell heat.

Why cooler air is risky: in part load the air going in was only about 28 °C colder than the stack and never below 557 °C. Option 3 needs a 300–550 °C gap, which can crack cells; the stack maker sets the allowed gap.

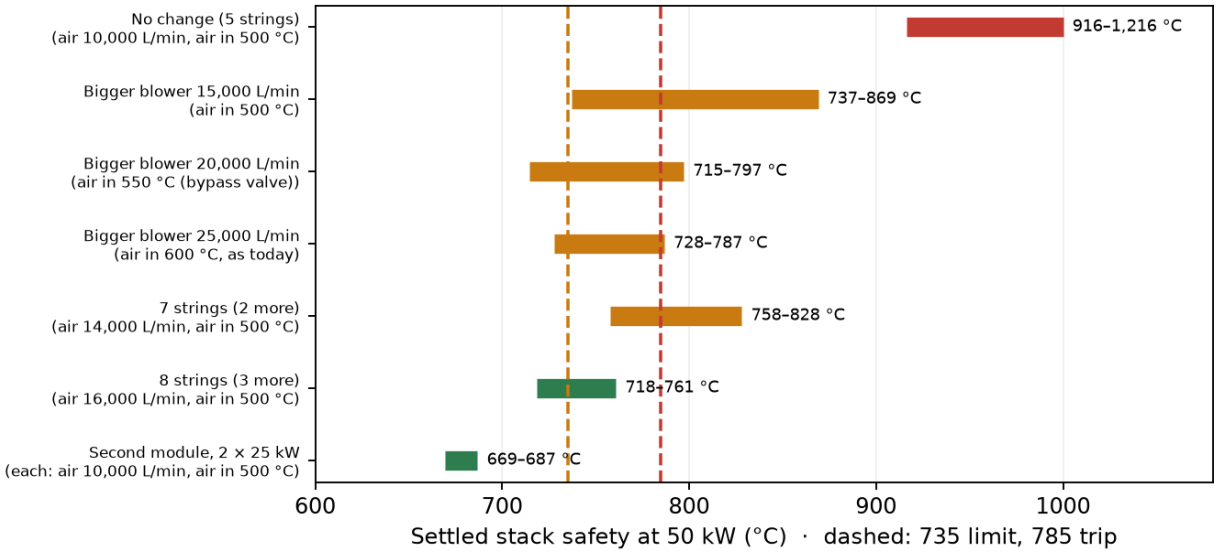
Not in the model: more internal reforming in the stack (takes in heat, cools from inside); better cells (higher voltage, less heat; today's cells already hit 0.72 V near 15 A); using more of the fuel (small effect).

Decisions: (1) set the rated power stage to about 33 kW (under 785 °C) or 29 kW (under 735 °C)? (2) ask the stack maker for air flow, coldest allowed air-in and internal reforming share at 50 kW; (3) short 20 A step test at air 10,000 L/min.

If you remember one thing: with today's equipment, lowering power to about 33–39 kW is the only change that settles under 785 °C; 50 kW needs about 2–3 times the cooling air.

5 Plant modifications for 50 kW

The surest fix is **more cells**: three more strings (8 in total) or a second module sharing the load. Each cell then gives less current, makes less heat and keeps a safe voltage. A **bigger air blower** alone can hold the temperature but leaves the cells at their 0.72 V lower limit.



Bar = range between the two heat rules. Green = under 785 °C in both rules, amber = one rule only, red = trips. More strings assume 10,000 L/min of air for every 5 strings.

Modification	Settles at	Cell voltage
No change (5 strings) (air 10,000 L/min, air in 500 °C)	916–1,216 °C	0.72 V (at limit)
Bigger blower 15,000 L/min (air in 500 °C)	737–869 °C	0.72 V (at limit)
Bigger blower 20,000 L/min (air in 550 °C (bypass valve))	715–797 °C	0.72 V (at limit)
Bigger blower 25,000 L/min (air in 600 °C, as today)	728–787 °C	0.72 V (at limit)
7 strings (2 more) (air 14,000 L/min, air in 500 °C)	758–828 °C	0.84 V
8 strings (3 more) (air 16,000 L/min, air in 500 °C)	718–761 °C	0.87 V
Second module, 2 × 25 kW (each: air 10,000 L/min, air in 500 °C)	669–687 °C	0.91 V

Not scored by the model: more reforming inside the stack (up to about 27 kW heat sink if all gas were reformed there, the same size as the 10–27 kW extra to remove; the stack maker must set the allowed share); better cells (+5 % voltage removes about 2.5 kW of 38.6 kW); higher limits (delay only, 15–35 min).

Every option also needs: after-burner and APH1/APH2 checked for more exhaust (300 °C trip); air heater (40 kW) and piping sized for more air; one 14 kW load per extra string; a rated-power temperature keeper (air first, then air-in, then current) acting before 735 °C.

Suggested order: (1) ask the stack maker for their 50 kW design (cells, air, coldest air-in, internal reforming, 735/785 °C); (2) test with no new hardware: 20 A step at 10,000 L/min, plus more internal reforming if they allow it; (3) choose more strings/module if voltage limits, or a bigger blower with bypass valve if 0.72 V is allowed.

If you remember one thing: 50 kW needs more cells sharing the load (fixes heat and voltage) or much more cooling air (fixes heat only).